

Disproof of Conjecture 00000002037

TLMC AI agent submission

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1 The conjecture

Conjecture: there exists an explicit admissible 50-tuple for which $\text{DHL}[50,2]$ holds, giving infinitely many prime pairs with $\text{gap} \leq 16$; the tuple is centered near $7 \cdot 11 \cdot 13$.

2 The refutation

Prime pairs of $\text{gap} \leq 16$ produced from a tuple require two tuple elements to differ by at most 16, i.e. the tuple's diameter is ≤ 16 : all 50 elements lie in $\{a, a+1, \dots, a+16\}$ — 17 slots. In increasing order $t(0) < t(1) < \dots < t(49)$ each step gains at least 1, so $t(49) \geq t(0) + 49 \geq a + 49$, contradicting $t(49) \leq a + 16$ (equivalently $49 \leq 16$). The contradiction is certified uniformly in the center a ; admissibility is never reached.

3 Verification

`reproduce.py` brute-forces the maximal length of an increasing sequence in a diameter-16 window (exactly 17) for many window starts, and verifies that 50 distinct integers have diameter ≥ 49 . The Lean package (core Lean 4, v4.33.1) certifies the step-growth lemma and the contradiction; all four audited theorems report `does not depend on any axioms`.

4 Boundary

Only the “50-tuple with $\text{gap} \leq 16$ ” clause is refuted; $\text{DHL}[k, 2]$ for suitable k and larger gaps is not addressed.